

Full length article

Effect of rescanning cycles on the characteristics of selective laser melting of Ti6Al4V

Zhongxu Xiao^a, Changpeng Chen^a, Zhiheng Hu^b, Haihong Zhu^{a,*}, Xiaoyan Zeng^a^a Wuhan National Laboratory for Optoelectronics, Huazhong University of Science and Technology, Wuhan, Hubei 430074, PR China^b Singapore Institute of Manufacturing Technology (SIMTech), Agency for Science, Technology and Research (A*STAR), 73 Nanyang Drive, Singapore 637662, Singapore

HIGHLIGHTS

- Relative density decreases when the sample is rescanned with too many cycles.
- Rescanning cycles don't change the microstructure of SLMed Ti6Al4V.
- The temperature and stress fields under different rescanning cycles are simulated.
- Mechanical properties of samples under different rescanning cycles are studied.

ARTICLE INFO

Keywords:

Rescanning cycles
Selective laser melting
Ti6Al4V
Mechanical properties
Residual stress
Microstructure

ABSTRACT

Rescanning has been introduced as a new method to improve the performance of selective laser melting (SLM) in recent years. However, many of the related works were done on the investigating of the rescanning parameters and only one rescanning cycle was used. In this work, the influences of rescanning cycle on the relative density, microstructure, residual stress and mechanical properties of SLMed Ti6Al4V alloy were investigated. The results show that as the rescanning cycles increase from 0 to 4, the relative density and residual stress of the samples increase at first and then decrease, the highest relative density and residual stress are found when the number of rescanning cycle is one. The number of rescanning cycle has little effect on the microstructure. The microstructures all consist of the martensitic α' laths which grow inside the columnar prior β grains. The ultimate tensile strength (UTS), yield strength, micro-hardness and micro-strain of the samples all increase as the rescanning cycles increase from 0 to 3, but decrease when the samples are rescanned fourth times. This research provides an insight in the influence of rescanning cycles on characteristics of SLM process and can be used to optimize the process parameters of SLM.

1. Introduction

As a promising additive manufacturing technique to fabricate high density and complex geometries metal parts, selective laser melting (SLM) develops rapidly in recent years [1,2]. A wide range of materials has been successfully processed with excellent mechanical properties [3–6]. However, high residual stress induced by the rapid cool rate and large thermal gradient is still a significant problem for SLM, which may lead to deformation and cracks [7].

In order to reduce the residual stress within the SLM components, many works have been done. Powder bed preheating is an effective method to decrease the temperature gradient and reduce the residual stress [8–10], but high preheating temperature is difficult to achieve in most commercial SLM machine [11]. Post heat treatment is often used

to reduce the residual stress and obtain the required mechanical properties, but this method will increase the manufacturing time and cost [11]. Rescanning method is rescanning the solidified layer during SLM. When the powder layer has been selectively fused, rescanning the solidified area will change the temperature gradient, which will have an effect on the stress during SLM. Compared to post heat treatment, rescanning method is also time consuming, but it can reduce the stress during the forming process. Usually, crack and warping caused by the stress during SLM will stop the whole forming process [12]. Furthermore, rescanning method is easy to be applied in most commercial SLM machine and the strategy can be changed easily. Therefore, rescanning has been regarded as a promising method to reduce the stress during the forming process in recent years.

In fact, there were some literatures related to this field. M. Shiomi

* Corresponding author.

E-mail address: zhuhh@mail.hust.edu.cn (H. Zhu).<https://doi.org/10.1016/j.optlastec.2019.105890>

Received 8 August 2019; Received in revised form 1 September 2019; Accepted 6 October 2019

Available online 12 October 2019

0030-3992/ © 2019 Elsevier Ltd. All rights reserved.

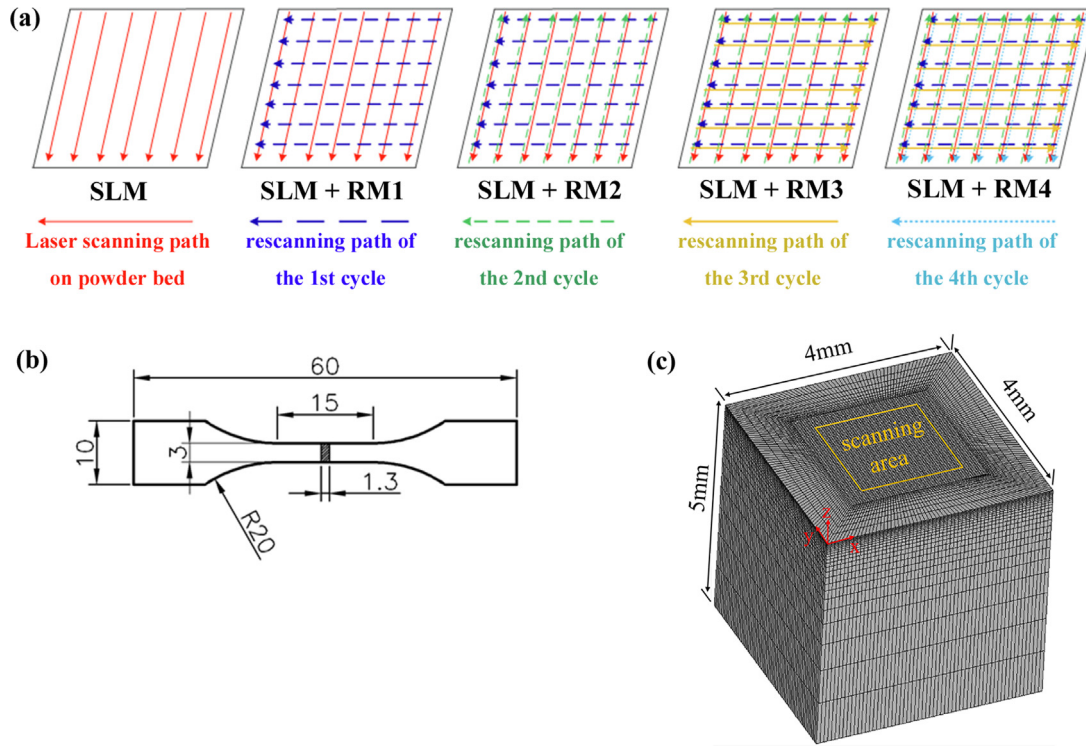


Fig. 1. Details of experiments and simulation: (a) scanning strategy (b) tensile sample (c) 3D finite element model and scanning region sketch.

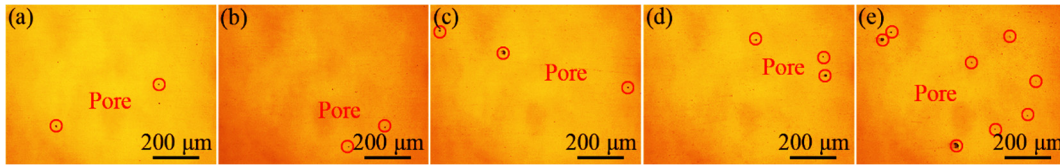


Fig. 2. OM images showing the pores of the samples under different rescanning conditions. (a) “SLM” sample, (b) “SLM + RM1” sample, (c) “SLM + RM2” sample, (d) “SLM + RM3” sample, (e) “SLM + RM4” sample.

et al. [8] used four different energy densities to rescan and found that the residual stress decreased with the energy density increase. The residual stress reduced 55% when the rescanning energy was 150% of the forming energy. J.P. Kruth et al. [13] rescanned SLMed Ti6Al4V parts with different scanning speed and indicated that only low scanning speed involving remelting could reduce the residual stress. High rescanning speed even led to the residual stress increasing. However, P. Mercelis et al. [14] pointed out that rescanning the parts with 50% of the forming energy resulted in a 30% reduction of the residual stress. When the part was rescanned with 100% of the forming energy, the residual stress didn't reduce. H. Ali et al. [15] studied the influence of rescanning on residual stress by varying the laser power and scanning speed in SLMed Ti6Al4V, respectively. They found that varying the scanning speed, the residual stress increased at first and then decreased, but varying the laser power, the trend of the residual stress was not clear. Scanning strategy has been proved important during SLM [16,17]. Correspondingly, C.P. Chen et al. [18] studied the influence of rescanning orientation on the rescanning and found that the residual stress was significantly reduced when using the vertical rescanning orientation.

After the samples were rescanned, not only the residual stress changed, but also the relative density, microstructure, mechanical properties and surface roughness changed. X.P. Li et al. [19] successfully fabricated crack-free Al85Ni5Y6Co2Fe2 bulk metallic glass composites by using a low power rescanning method. E. Yasa et al. [20] rescanned the 316L stainless steel and found that the residual porosity decreased from 0.77% to 0.036% and the roughness of the top surface

decreased from 12 μm to 1.5 μm when the laser rescanning was applied with appropriate parameters. Y. Tian et al. [21] studied the influence of the laser rescanning on the electron beam melted (EBM) Ti6Al4V component. They found that the surface roughness reduced 75% and the hardness had a small increase. In addition, the thin surface layer remelted exhibited a re-orientated texture. S. Griffiths et al. [22] found that rescanning the Al-Mg-Zr alloy during SLM led to a columnar to equiaxed microstructural transition. H. Ali et al. [15] pointed out that the microstructure of the Ti6Al4V didn't change and the mechanical properties variation tendency was unsure when the rescanning speed and laser power increased.

However, most researchers only paid attention to one rescanning cycle. There was few literatures related to the effect of the rescanning cycles. K. Wei et al. [23] found that the number of rescanning cycles had a significant effect on the residual stress and mechanical properties of SLMed Ti-5Al-2.5Sn. As the rescanning cycles increased, the residual stress of the SLMed Ti-5Al-2.5Sn samples increased first and then decreased, while the elongation slightly increased and surface roughness decreased. As a widely used alloy in aerospace, automotive and medical sectors [24], SLMed Ti6Al4V has become attractive [25]. There are many literatures reported SLMed Ti6Al4V, including process parameters optimization [26], heat treatment [27], thermal stress [18,28] and dimensional accuracy [29]. Now, SLMed Ti6Al4V has been successful and utilized in industry. However, the influence of the rescanning cycles on the characteristics of SLMed Ti6Al4V was still absent. In this paper, SLMed Ti6Al4V samples rescanned from 0 to 4 cycles were fabricated. The changes in relative density, microstructure, residual

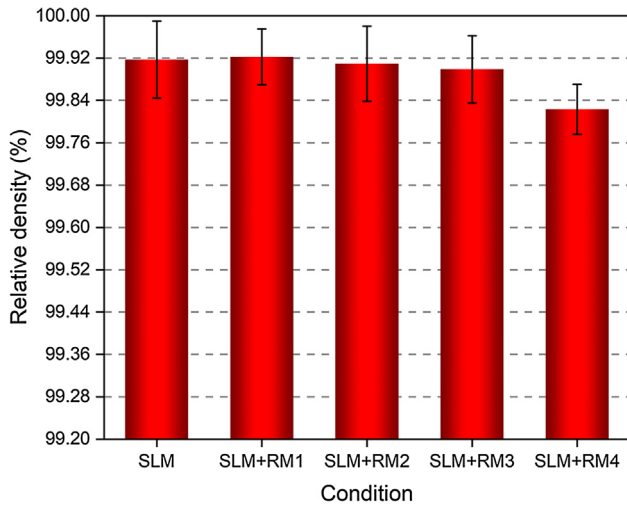


Fig. 3. Relative density of the samples under different rescaning conditions.

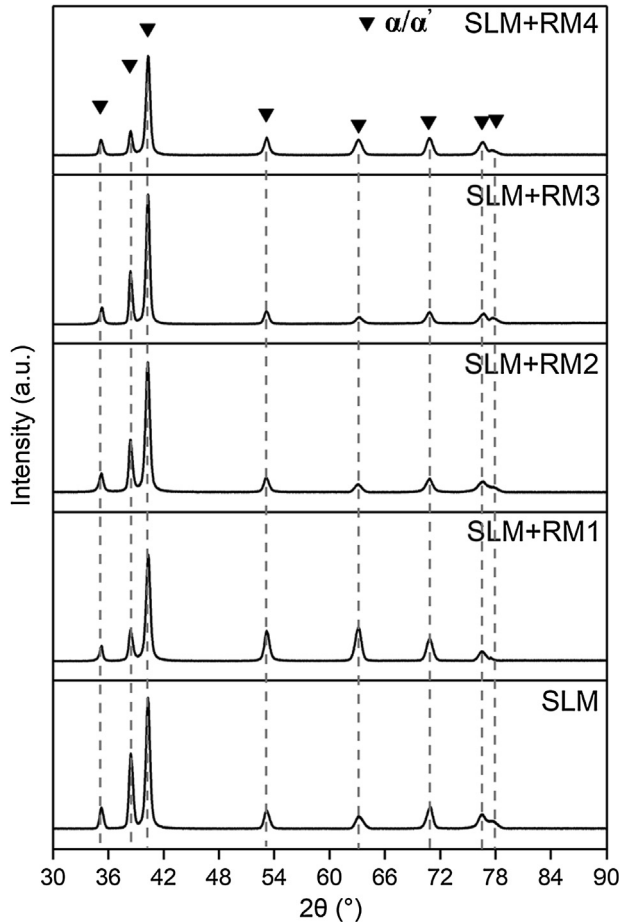


Fig. 4. X-ray diffraction patterns of the SLM samples under different rescaning conditions.

stress and mechanical properties were systematically investigated. Accordingly, the mechanism of all the changes was understood. The results would bring insights into the full-scale influence of the rescaning cycles during SLM.

2. Experimentals and numerical models

Gas-atomized Ti6Al4V powder with the size of 20–50 μm was used

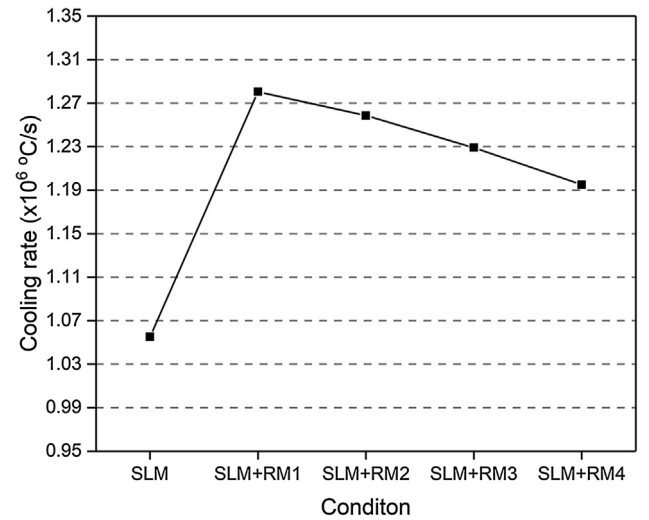


Fig. 5. The simulated cooling rate of the samples under different rescaning conditions.

in this work. The experiments were conducted on a self-developed SLM machine LSNF-1 whose details were introduced in our previous works [30]. Cuboid samples in the size of $15 \times 15 \times 5 \text{ mm}^3$ were fabricated with the process parameters of laser power 200 W, scanning speed 1000 mm/s, hatch spacing 0.08 mm and layer thickness 0.02 mm. The phase angle is 90° . All samples were built in the argon atmosphere with the O_2 concentration below 200 ppm. The rescaning process parameters were the same as the building process parameters, but scanning direction was perpendicular with the previous scanning direction, as shown in Fig. 1a. For the convenience of description, samples without rescaning were defined as “SLM”, samples rescanned 1, 2, 3 and 4 cycles were defined as “SLM + RM1”, “SLM + RM2”, “SLM + RM3”, and “SLM + RM4” in the following analyses, respectively.

The relative density was the average value which was evaluated by image processing of eight cross-sectional optical micrographs using the Image-Pro Plus 6.0 software. The microstructure was characterized using a Nikon Epiphot-300 optical microscope (OM), Zeiss-EVO-18 scanning electron microscope (SEM) and an X'Pert-Pro X-ray diffractometer (XRD) using $\text{Cu K}\alpha$ ($\lambda = 0.15406 \text{ nm}$) radiation. Micro-hardness tests were conducted with an HVS-1000 tester at a load of 500 g and a holding time of 20 s. The residual stress at the middle of the top surface of the sample was measured using the hole-drilling method (HDM) according to ASTM: E837-08e2 standard. The diameter of the drilling hole was 1.5 mm and the depth was 2 mm. Thus, the measured residual stress is the average residual stress of the whole hole. The shape of the tensile specimens built with the same rescaning conditions is shown in Fig. 1b. Tensile tests were performed using the Zwick/Roell tester at room temperature with a rate of 2 mm/min. Three samples were tested for each condition.

The micro-strain was calculated by the jade 6 software, using the Williamson-Hall (WH) plots [31]. It can be calculated by Eq. (1):

$$\beta \cos \theta = \frac{K\lambda}{D} + 4\varepsilon \tan \theta \quad (1)$$

where β is the broadening of each diffraction peak, K is the shape factor, λ is the wavelength of $\text{Cu K}\alpha$, θ is the angle of each diffraction peak. D is the crystalline size and the ε is the micro-strain.

The details of the finite element model for the SLM process has been comprehensively validated both in thermal analysis and residual stress analysis in our previous work [18], i.e., the molten pool dimension predicted by the thermal analysis, residual stress distribution and variation simulated by the mechanical analysis were in good agreement with the experimental data. Therefore, the validation process provided confidence to apply the developed computational technique to other

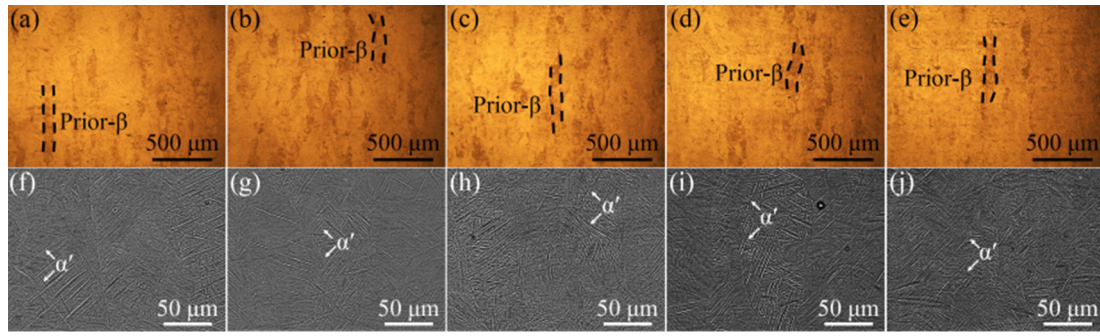


Fig. 6. OM and SEM images of the “SLM” sample (a, f), “SLM + RM1” sample (b, g), “SLM + RM2” sample (c, h), “SLM + RM3” sample (d, i), “SLM + RM4” sample (e, j).

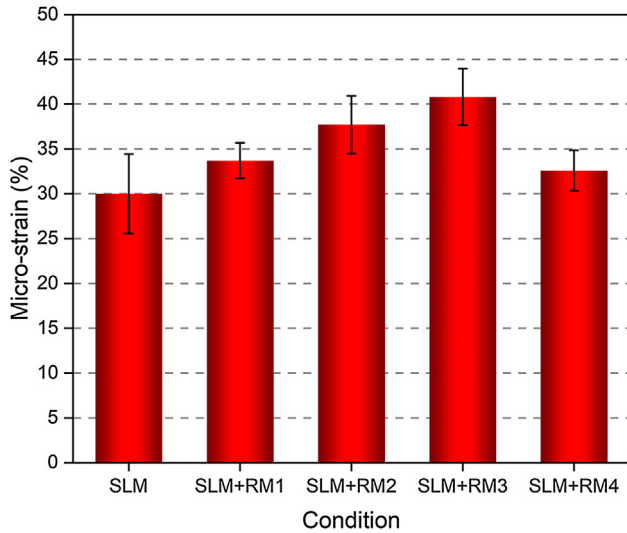


Fig. 7. Micro-Strain of samples under different rescanning conditions.

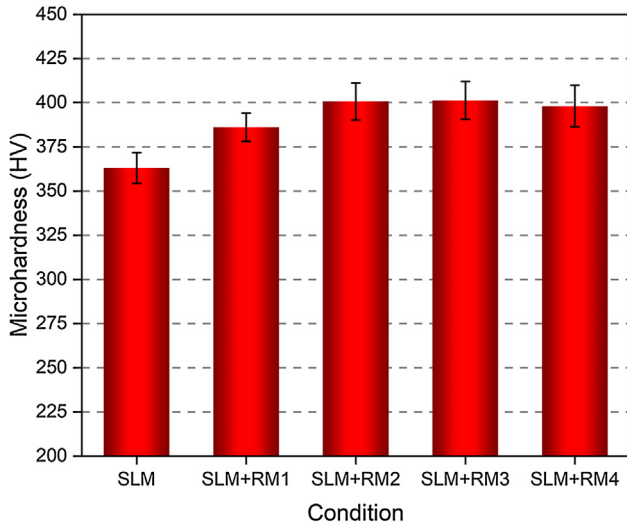


Fig. 8. Micro-hardness of the samples under different rescanning conditions.

additional applications involving additive manufacturing study. On the basis of the above, the effect of the rescanning cycles on the residual stress in SLM was carried out. As shown in Fig. 1c, the residual stress of the single layer with a size of $4 \times 4 \times 0.02 \text{ mm}^3$ was simulated. The laser scanning area was $1.2 \times 1.2 \text{ mm}^2$. The substrate size was $4 \times 4 \times 5 \text{ mm}^3$.

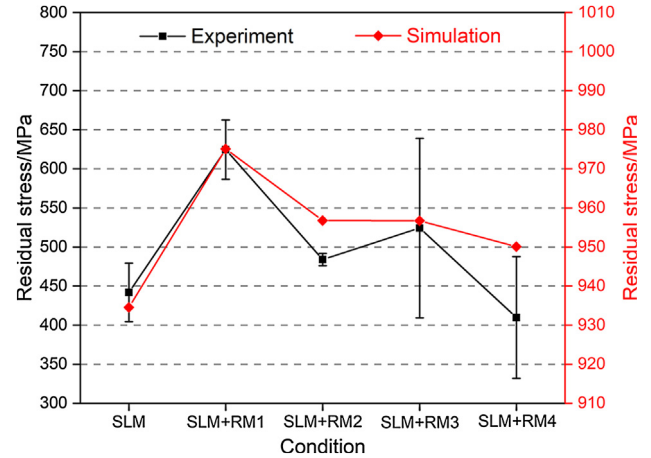


Fig. 9. Residual stress of samples under different rescanning conditions.

3. Results and discussion

3.1. Relative density

The OM image and relative density of the samples under different rescanning conditions are shown in Figs. 2 and 3, respectively. It can be found that there are some pores in the samples. The relative density increases at first and then decreases with the rescanning cycles increasing. The pores are almost all round pores. The highest relative density is the “SLM + RM1” samples, whose value is 99.92%. That's because increasing the rescanning cycle can prolong the existence of the molten pool, therefore contributing to the gas escaping as well as high-quality metallurgical bonding between adjacent tracks and layers [23,32]. When the rescanning cycle is too much, the heat from the melt pool is not conducted into the surrounding material fast enough and the melt pool starts to evaporate. Thereby, the molten droplets spatter out from the boiling melt pool and generate the micro porosity [32].

3.2. Microstructure

The XRD patterns of the Ti6Al4V samples under different rescanning conditions are shown in Fig. 4. Ti6Al4V is an $\alpha + \beta$ alloy. However, the XRD results indicate that the SLMed Ti6Al4V samples consist of the Hexagonal Close Packed (HCP) α/α' -Ti hexagonal. There are no Body-Centered Cubic (BCC) structured β phase diffraction peaks in the XRD patterns of Fig. 4. This is because the cooling rate of SLM is very high [33]. For Ti6Al4V alloy, cooling rate higher than 410°C/s leads to fully martensitic microstructure α' [34]. The cooling rates of SLMed Ti6Al4V under different rescanning cycles are shown in Fig. 5, it can be found that the cooling rates under different rescanning cycles are all much higher than 410°C/s , so the microstructures of the all samples only

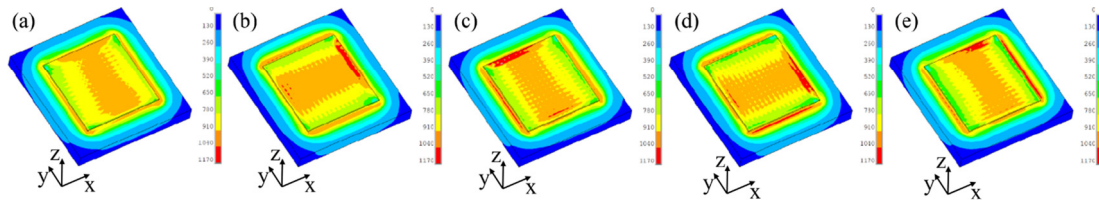


Fig. 10. The equivalent stress distribution of the single layer under different rescanning conditions: (a) “SLM” sample, (b) “SLM + RM1” sample, (c) “SLM + RM2” sample, (d) “SLM + RM3” sample, (e) “SLM + RM4”.

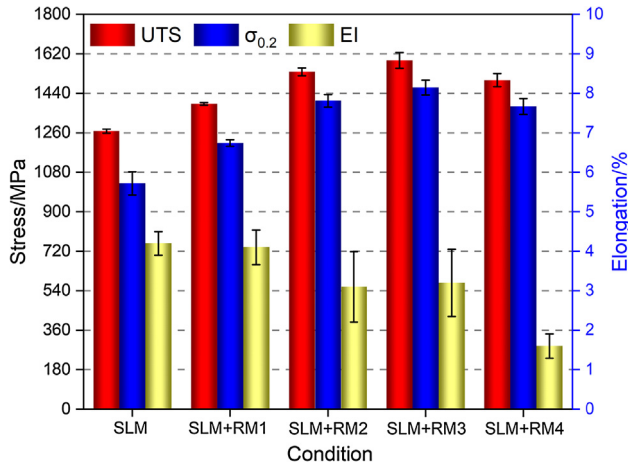


Fig. 11. Tensile properties of samples under different rescanning conditions.

consist of martensitic microstructure α' .

In addition, with the rescanning cycles increasing, the cooling rate increases at first and then decreases, the highest cooling rate is the SLM + RM1 sample of 1.28×10^6 °C/s. This is because when sample is fabricated without rescanning, the laser scanning path is surrounded by powder, when the samples are rescanned, the laser scans on the metal part and is surrounded by solid. The thermal conductivity of the metal powder is much lower than that of the metal solid [18]. Therefore, the cooling rates of the rescanning samples are all higher than that of the sample without rescanning. When the rescanning cycles increase, the temperature of the metal part increases, so the cooling rate decreases.

The OM and SEM images of the Ti6Al4V sample under different rescanning conditions are shown in Fig. 6. From Fig. 6ae, it can be found that for all the samples, there are no evidence difference. The length of the columnar prior β grains reaches the scale of millimeter, which means that the epitaxial growth of the columnar prior β grains penetrates through dozens of the layers. This is because the localized heat inputs lead to the directional solidification during SLM. Fig. 6f-j is the SEM images of the Ti6Al4V samples under different rescanning conditions. It can be found that the martensitic α' laths growing inside

the columnar prior β grains. During solidification of SLM, the β phase preferentially grows in the $\langle 100 \rangle$ direction which is parallel to the build direction. Due to the fast cooling, the β phase then transforms into the α' phase. According to the Burgers relation, the $(0001)_{\alpha'}$ plane of each α' lath is parallel to one of the $(110)_{\beta}$ plane and $\langle 11\bar{2}0 \rangle_{\alpha'}$ is parallel to the $\langle 111 \rangle_{\beta}$, therefore, the α' laths is about 45° to the columnar prior β grains boundaries [35,36].

3.3. Micro-hardness and Micro-strain

The micro-strain inside the “SLM” samples, “SLM + RM1” samples, “SLM + RM2” samples, “SLM + RM3” samples and “SLM + RM4” samples calculated through the XRD patterns are shown in the Fig. 7. It can be found that with the rescanning cycles increasing from 0 to 3, the micro-strain inside the samples increases. There are many thermal cycles during SLM [37,38], when the laser beam irradiates the surface, these thermal cycles result in the stress/strain cycles [39]. When the rescanning method is utilized, the strain field inside the Ti6Al4V samples is affected. The strain cycles caused by the thermal cycles result in the strain hardening in the Ti6Al4V samples. So the micro-strain of the sample increases. When the sample is rescanned 4 cycles, the sample was annealed and the micro-strain in the “SLM + RM4” sample decreases as shown in Fig. 7.

The micro-hardness of the samples under different rescanning conditions is shown in Fig. 8. It can be found that the micro-hardness increases first and then decreases when the rescanning cycles increase from 0 to 4 times. The highest micro-hardness is 401.28HV in the “SLM + RM3” sample. The variation of the micro-hardness agrees well with the variation of the micro-strain. It further indicates that the rescanning method can lead to the strain hardening in the Ti6Al4V samples and the sample is annealed when it is rescanned four cycles.

3.4. Residual stress

Fig. 9 shows the residual stress of the Ti6Al4V samples under different rescanning conditions. Different from the study of K. Wei et al. [19], it can be found that the residual stresses of the rescanned samples are higher than those of the samples without rescanning, especially for “SLM + RM1” samples, which is more than 39% compared to the

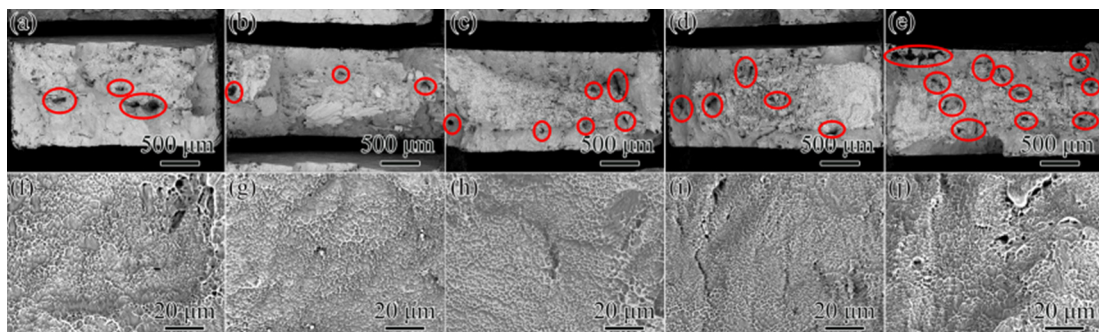


Fig. 12. Tensile fracture images of the “SLM” sample (a, f), “SLM + RM1” sample (b, g), “SLM + RM2” sample (c, h), “SLM + RM3” sample (d, i), “SLM + RM4” sample (e, j).

“SLM” sample. This result agrees well with the cooling rate result shown in Fig. 5. When the laser beam rescanning the samples at the first cycle (“SLM + RM1”), the laser scans on the metal solid, whose thermal conductivity is much higher than that of the powder. As a result, the cooling rate and thermal gradient of the “SLM + RM1” sample are relatively higher than that of the “SLM” sample. When the samples are rescanned with more cycles, the heating accumulation effect increases the initial temperature of the metal and decreases the cooling rate as well as the thermal gradient. Therefore, the residual stress reduces correspondingly. The simulation results also show similar trend. It is worthy to know, porosity also influences the residual stress. However, T Mishurova et al. proved that the influence of the porosity on the residual stress could be neglected if the porosity is lower than 1% [2]. According to Fig. 3, the relative density of all samples are higher than 99.8%. Therefore, the influence of the porosity on the residual stress can be neglected.

The equivalent stress distribution of the single layer under different rescanning conditions is shown in Fig. 10. It can be found that the high-stress region increases at first and then decreases with the rescanning cycles increasing from 0 to 4. The biggest high-stress region appears in the “SLM + RM1” sample. The stress distribution of the “SLM” samples is the most uniform. These results all related to the cooling rate variation tendency [15]. The lower the cooling rate is, the more uniform the temperature field is, and the smaller the residual stress is.

3.5. Mechanical properties

Tensile properties of the samples experiencing different rescanning cycles are shown in Fig. 11. It can be seen that the application of rescanning is an effective way to improve both the ultimate tensile strength (UTS) and the yield strength of the samples. The UTS and yield strength of the samples increase from 1268 MPa, 1030 MPa of the “SLM” sample to 1392 MPa, 1213 MPa of the “SLM + RM1”, 1538 MPa, 1406 MPa of “SLM + RM2”, 1590 MPa, 1466 MPa of “SLM + RM3” and 1500 MPa, 1380 MPa of “SLM + RM4”, respectively. As the rescanning cycle increases, the UTS and yield strength of the samples increases first and then decreases. The maximum UTS and yield strength are the samples experiencing 3 rescanning cycles. However, the elongations of the rescanned samples are all lower than that of the “SLM” samples. The elongation decreases along with the rescanning cycles.

The strengthening mechanism of Ti6Al4V is the strain field induced by the martensite and the dislocation density inside the laths [40]. As shown in Fig. 7, there is strain strength in the rescanning samples, so the UTS and yield strength of the rescanning samples are larger than that of the “SLM” samples, and the elongations of the rescanning samples are smaller than that of the “SLM” sample. Furthermore, as the rescanning cycles increase, the micro-strain increases first and then decreases, and the “SLM + RM3” sample shows the highest micro-strain. Therefore, the UTS and yield strength of the rescanning samples increase first and then decrease with the rescanning cycles increasing, and the “SLM + RM3” sample shows the highest UTS and yield strength. The elongation should decrease with the rescanning cycles increasing from 1 to 3 and increase in “SLM + RM4” sample. However, the relative density of “SLM + RM4” sample is only 99.82% and there are large amounts of porosity in the sample as shown in Fig. 12e, therefore, “SLM + RM4” sample shows low UTS, yield strength and elongation. H. Ali et al. [15] also found that rescanning the SLMed Ti64 sample with high energy density resulted in the reduction of the elongation because of the increase of the porosity.

Fig. 12 shows the tensile fracture images of the samples under different rescanning conditions. From Fig. 12a–e, it can be found that the variation tendency of the defects is just opposite to the variation tendency of the relative density discussed in Section 3.1. The defects decrease at first and then increase with the rescanning cycles increasing. Large amounts of the defects lead to the reduction of the elongation of “SLM + RM4” sample.

Fig. 12f–j shows the dimples of the tensile samples. It indicates that all the samples are ductile fracture. In addition, it can be found that with the rescanning cycles increasing, the size of the dimples decreases at first and then increases. It further indicates that the rescanning method can lead to the strain hardening in the Ti6Al4V samples and the “SLM + RM4” sample is annealed.

4. Conclusions

In this paper, Ti6Al4V samples rescanned from 0 to 4 cycles with same process parameters were fabricated and the influence of the rescanning cycles on the relative density, microstructure, residual stress and mechanical properties of the Ti6Al4V alloy were systematically investigated. Following conclusions can be drawn:

- (1) The relative density of the samples increases at first and then decreases with the rescanning cycles increases from 0 to 4. The highest relative density of 99.92% is found for the “SLM + RM1” sample. When the samples are rescanned 4 cycles, the relative density decreases to 99.82%. This is caused by the preheating effect of the multi laser scanning, inducing the temperature and the life time of the molten pool increase and the porosity decreases. However, when the rescanning cycles are large, evaporation occurs because of the extremely high temperature of the molten pool, which induces the porosity increases.
- (2) The residual stress of the “SLM + RM1” samples is higher than that of the samples without rescanning. The residual stress of the “SLM + RM1” sample exceeds 39% residual stress of the “SLM” sample. However, the residual stress decreases as the rescanning cycles increases. This is because the rescanning happens on the solid metal which thermal conductivity is much higher than that of the metal powder, inducing the cooling rate increases. As the rescanning cycle increases, cooling rate decreases and residual stress decreases too.
- (3) The rescanning cycles have no evident effect on the microstructure. The microstructure consists of the martensitic α' laths growing inside the columnar prior β grains regardless of the rescanning cycles. The growing direction of the martensitic α' laths is about 45° to the columnar prior β grains boundary and the martensitic α' laths are mutually perpendicular.
- (4) The ultimate tensile strength (UTS), yield strength, micro-hardness and micro-strain of the samples all increase with the rescanning cycles increasing from 0 to 3, but decrease when the sample is rescanned 4 times. This is resulted from the integrated effect of the annealing and strain strength of rescanning. The elongation of the samples under different rescanning cycles is all lower than that of the sample without rescanning.

Acknowledgements

This research is supported by the National Program on Key Basic Research Project of China (613281) and the human spaceflight program of China (D050302). The authors also thank the Analytical and Testing Center of HUST for the SEM and XRD analysis.

References

- [1] G.M. Bilgin, Z. Esen, Ş.K. Akın, A.F. Dericioglu, Optimization of the mechanical properties of Ti-6Al-4V alloy fabricated by selective laser melting using thermo-hydrogen processes, *Mater. Sci. Eng., A* 700 (2017) 574–582.
- [2] T. Mishurova, K. Artzt, J. Haubrich, G. Requena, G. Bruno, New aspects about the search for the most relevant parameters optimizing SLM materials, *Addit. Manuf.* 25 (2019) 325–334.
- [3] K. Kunze, T. Etter, J. Grässlin, V. Shklover, Texture, anisotropy in microstructure and mechanical properties of IN738LC alloy processed by selective laser melting (SLM), *Mater. Sci. Eng., A* 620 (2015) 213–222.
- [4] M. Strantz, B. Vrancken, M.B. Prime, C. Truman, M. Rombouts, D.W. Brown, P. Guillaume, D. Van Hemelrijck, Directional and oscillating residual stress on the

- mesoscale in additively manufactured Ti-6Al-4V, *Acta Materialia* (2019).
- [5] M. Tang, P.C. Pistorius, J.L. Beuth, Prediction of lack-of-fusion porosity for powder bed fusion, *Addit. Manuf.* 14 (2017) 39–48.
 - [6] H. Liao, H. Zhu, G. Xue, X. Zeng, Alumina loss mechanism of Al₂O₃-AlSi10 Mg composites during selective laser melting, *J. Alloys Compd.* 785 (2019) 286–295.
 - [7] H. Shipley, D. McDonnell, M. Culleton, R. Coull, R. Lupoi, G. O'Donnell, D. Trimble, Optimisation of process parameters to address fundamental challenges during selective laser melting of Ti-6Al-4V: a review, *Int. J. Mach. Tool Manu.* 128 (2018) 1–20.
 - [8] M. Shiomi, K. Osakada, K. Nakamura, T. Yamashita, F. Abe, Residual stress within metallic model made by selective laser melting process, *CIRP Ann.* 53 (1) (2004) 195–198.
 - [9] S. Le Roux, M. Salem, A. Hor, Improvement of the bridge curvature method to assess residual stresses in selective laser melting, *Addit. Manuf.* 22 (2018) 320–329.
 - [10] L.-E. Loh, C.-K. Chua, W.-Y. Yeong, J. Song, M. Mapar, S.-L. Sing, Z.-H. Liu, D.-Q. Zhang, Numerical investigation and an effective modelling on the Selective Laser Melting (SLM) process with aluminium alloy 6061, *Int. J. Heat Mass Transf.* 80 (2015) 288–300.
 - [11] H. Ali, L. Ma, H. Ghadbeigi, K. Mumtaz, In-situ residual stress reduction, martensitic decomposition and mechanical properties enhancement through high temperature powder bed pre-heating of Selective Laser Melted Ti6Al4V, *Mater. Sci. Eng., A* 695 (2017) 211–220.
 - [12] D. Wang, Y. Yang, Z. Yi, X. Su, Research on the fabricating quality optimization of the overhanging surface in SLM process, *Int. J. Adv. Manuf. Technol.* 65 (9) (2013) 1471–1484.
 - [13] J.-P. Kruth, J. Deckers, E. Yasa, R. Wauthlé, Assessing and comparing influencing factors of residual stresses in selective laser melting using a novel analysis method, *Proc. Inst. Mech. Eng., Part B: J. Eng. Manuf.* 226 (6) (2012) 980–991.
 - [14] P. Mercelis, J.P. Kruth, Residual stresses in selective laser sintering and selective laser melting, *Rapid Prototyping J.* 12 (5) (2006) 254–265.
 - [15] H. Ali, H. Ghadbeigi, K. Mumtaz, Effect of scanning strategies on residual stress and mechanical properties of Selective Laser Melted Ti6Al4V, *Mater. Sci. Eng., A* 712 (2018) 175–187.
 - [16] N.T. Aboulkhair, N.M. Everitt, I. Ashcroft, C. Tuck, Reducing porosity in AlSi10Mg parts processed by selective laser melting, *Addit. Manuf.* 1–4 (2014) 77–86.
 - [17] B. Cheng, S. Shrestha, K. Chou, Stress and deformation evaluations of scanning strategy effect in selective laser melting, *Addit. Manuf.* 12 (2016) 240–251.
 - [18] C. Chen, J. Yin, H. Zhu, Z. Xiao, L. Zhang, X. Zeng, Effect of overlap rate and pattern on residual stress in selective laser melting, *Int. J. Mach. Tools Manuf.* 145 (2019) 103433.
 - [19] X.P. Li, C.W. Kang, H. Huang, T.B. Sercombe, The role of a low-energy-density rescanning in fabricating crack-free Al85Ni5Y6Co2Fe2 bulk metallic glass composites via selective laser melting, *Mater. Des.* 63 (2014) 407–411.
 - [20] E. Yasa, J.P. Kruth, J. Deckers, Manufacturing by combining Selective Laser Melting and Selective Laser Erosion/laser re-melting, *CIRP Ann.* 60 (1) (2011) 263–266.
 - [21] Y. Tian, W.S. Gora, A.P. Cabo, L.L. Parimi, D.P. Hand, S. Tammas-Williams, P.B. Prangnell, Material interactions in laser polishing powder bed additive manufactured Ti6Al4V components, *Addit. Manuf.* 20 (2018) 11–22.
 - [22] S. Griffiths, M.D. Rossell, J. Croteau, N.Q. Vo, D.C. Dunand, C. Leinenbach, Effect of laser rescanning on the grain microstructure of a selective laser melted Al-Mg-Zr alloy, *Mater. Charact.* 143 (2018) 34–42.
 - [23] K. Wei, M. Lv, X. Zeng, Z. Xiao, G. Huang, M. Liu, J. Deng, Effect of laser remelting on deposition quality, residual stress, microstructure, and mechanical property of selective laser melting processed Ti-5Al-2.5Sn alloy, *Mater. Charact.* 150 (2019) 67–77.
 - [24] J.J. Lin, Y.H. Lv, Y.X. Liu, B.S. Xu, Z. Sun, Z.G. Li, Y.X. Wu, Microstructural evolution and mechanical properties of Ti-6Al-4V wall deposited by pulsed plasma arc additive manufacturing, *Mater. Des.* 102 (2016) 30–40.
 - [25] J.Y. Cho, W. Xu, M. Brandt, M. Qian, Selective laser melting-fabricated Ti-6Al-4V alloy: microstructural inhomogeneity, consequent variations in elastic modulus and implications, *Opt. Laser Technol.* 111 (2019) 664–670.
 - [26] A. Khorasani, I. Gibson, U.S. Awan, A. Ghaderi, The effect of SLM process parameters on density, hardness, tensile strength and surface quality of Ti-6Al-4V, *Addit. Manuf.* 25 (2019) 176–186.
 - [27] P. Kumar, U. Ramamurty, Microstructural optimization through heat treatment for enhancing the fracture toughness and fatigue crack growth resistance of selective laser melted Ti6Al4V alloy, *Acta Materialia* 169 (2019) 45–59.
 - [28] P. Tan, F. Shen, B. Li, K. Zhou, A thermo-metallurgical-mechanical model for selective laser melting of Ti6Al4V, *Mater. Des.* 168 (2019) 107642.
 - [29] L. Zhang, S. Zhang, H. Zhu, Z. Hu, G. Wang, X. Zeng, Horizontal dimensional accuracy prediction of selective laser melting, *Mater. Des.* 160 (2018) 9–20.
 - [30] Z. Hu, H. Zhang, H. Zhu, Z. Xiao, X. Nie, X. Zeng, Microstructure, mechanical properties and strengthening mechanisms of AlCu5MnCdVA aluminum alloy fabricated by selective laser melting, *Mater. Sci. Eng., A* 759 (2019) 154–166.
 - [31] J. Pelleg, E. Elsh, D. Mogilyanski, Evaluation of average domain size and microstrain in a silicide film by the Williamson-Hall method, *Metall. Mater. Trans. A* 36 (11) (2005) 3187–3194.
 - [32] S. Pal, G. Lojen, V. Kokol, I. Drstvensek, Evolution of metallurgical properties of Ti-6Al-4V alloy fabricated in different energy densities in the Selective Laser Melting technique, *J. Manuf. Process.* 35 (2018) 538–546.
 - [33] H.K. Rafi, N.V. Karthik, H. Gong, T.L. Starr, B.E. Stucker, Microstructures and mechanical properties of Ti6Al4V parts fabricated by selective laser melting and electron beam melting, *J. Mater. Eng. Perform.* 22 (12) (2013) 3872–3883.
 - [34] T. Ahmed, H.J. Rack, Phase transformations during cooling in $\alpha + \beta$ titanium alloys, *Mater. Sci. Eng., A* 243 (1) (1998) 206–211.
 - [35] B. Vrancken, L. Thijs, J.-P. Kruth, J. Van Humbeeck, Heat treatment of Ti6Al4V produced by Selective Laser Melting: microstructure and mechanical properties, *J. Alloys Compd.* 541 (2012) 177–185.
 - [36] C. Qiu, N.J.E. Adkins, M.M. Attallah, Microstructure and tensile properties of selectively laser-melted and of HIPed laser-melted Ti-6Al-4V, *Mater. Sci. Eng., A* 578 (2013) 230–239.
 - [37] L. Parry, I.A. Ashcroft, R.D. Wildman, Understanding the effect of laser scan strategy on residual stress in selective laser melting through thermo-mechanical simulation, *Addit. Manuf.* 12 (2016) 1–15.
 - [38] N.E. Hodge, R.M. Ferencz, J.M. Solberg, Implementation of a thermomechanical model for the simulation of selective laser melting, *Comput. Mech.* 54 (1) (2014) 33–51.
 - [39] L. van Belle, G. Vansteenkiste, J.C. Boyer, Investigation of residual stresses induced during the selective laser melting process, *Key Eng. Mater.* 554–557 (2013) 1828–1834.
 - [40] H.K. Rafi, T.L. Starr, B.E. Stucker, A comparison of the tensile, fatigue, and fracture behavior of Ti-6Al-4V and 15-5 PH stainless steel parts made by selective laser melting, *Int. J. Adv. Manuf. Technol.* 69 (5) (2013) 1299–1309.